PHY302: EndSem Examination [50 points]

Instructor: Manabendra Nath Bera

Negative markings:
(a) Missing name and/or registration number on the answer sheet. [–3 points]
(b) Writing that is difficult to read or follow. [–3 points]

1. Indicate **T** or **F** for true or false, respectively, with clear justification. [10 points]

(a) Consider any operator A and two arbitrary complete sets of orthonormal bases $\{|\psi_i\rangle\}$ and $\{|\phi_m\rangle\}$, where $\langle\psi_i|\phi_m\rangle \neq 0$. Then, $\sum_i \langle\psi_i|A|\psi_i\rangle \neq \sum_m \langle\phi_m|A|\phi_m\rangle$. 1

(b) The uncertainty ΔS of a time-independent operator S is time-independent.

(c) For any finite integer $N \geq 2$, there exist $(N \times N)$-matrices, X and P, such that $[X, P] = i\,\mathbb{I}$. 0

(d) Let $T \otimes S$ be a linear operator on the vector space $\mathcal{V} \otimes \mathcal{W}$. Then $(T \otimes S)^\dagger = S^\dagger \otimes T^\dagger$. 1

(e) $\hat{\mathbf{r}} \cdot \hat{\mathbf{p}} = \hat{\mathbf{p}} \cdot \hat{\mathbf{r}} - 3i\hbar\mathbb{I}$. Here $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ are the position and momentum operators in 3-dimension, respectively.

(f) For a simple harmonic oscillator the number operator in the Heisenberg picture is identical to the number operator in Schrödinger picture.

(g) $[\mathbf{L}, \hat{x}^2] = 0$. Here $\mathbf{L}$ is orbital angular momentum operator, and $\hat{x}$ is the position-x operator.

(h) If $\mathbf{J}_1$ and $\mathbf{J}_2$ are sets of operators each satisfying the algebra of angular momentum, the combination $\mathbf{J}_1 - \mathbf{J}_2$ also satisfies the algebra of angular momentum.

(i) Let $\mathbf{J}_1$ and $\mathbf{J}_2$ be angular momentum operators. The operator $\mathbf{J}_1 \cdot \mathbf{J}_2$ commutes with $J_{1x} + J_{2x}$.

(j) The wave function of a particle in a three-dimensional central potential must vanish at the origin.

2. (a) Consider the one-dimensional time-independent Schrödinger equation for some arbitrary potential $V(x)$. Prove that if a solution $\psi(x)$ has the property that $\psi(x) \to 0$ as $|x| \to \infty$, then the solution must be non-degenerate and real, apart from a possible overall phase factor. [3+3 points] 6

(b) A particle of mass m moves in one dimension under the influence of a potential $V(x)$. Suppose it is in an energy eigenstate $\psi(x) = (\gamma^2/\pi)^{1/4} \exp\left(-\gamma^2 x^2/2\right)$ with energy $E = \hbar^2\gamma^2/2m$.

(i) Find the mean position of the particle. [1 point]

(ii) Find the mean momentum of the particle. [1 point]

(iii) Find $V(x)$. [2 points]

3. Consider a one-dimensional simple harmonic oscillator. Do the following algebraically, i.e., without using the explicit forms of the wave-functions.

(a) Construct a state as a linear combination of $|0\rangle$ and $|1\rangle$ such that $\langle\hat{x}\rangle$ is as large as possible. Here $|0\rangle$ and $|1\rangle$ are the ground and first excited states. [3 points]

(b) Suppose the oscillator is in the state constructed in (i) at time $t = 0$. What is the state vector for $t > 0$? [1 point]

(c) Evaluate the expectation value $\langle \hat{x} \rangle$ as a function of time for $t > 0$ using (i) the Schrödinger picture and (ii) the Heisenberg picture independently. [3 points] same

(d) Evaluate $\langle (\Delta x)^2 \rangle$ as a function of time using both Schrödinger and Heisenberg pictures. [3 points]

Note, $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)$, *where* $a^\dagger$ *is the creation operator.*

4. (a) A state $|l, m\rangle$ is a simultaneous eigenstate of the orbital angular momentum operators $\mathbf{L}^2$ and L_z:

$$\mathbf{L}^2|l, m\rangle = \hbar^2\, l(l+1)\, |l, m\rangle, \quad L_z|l, m\rangle = \hbar\, m\, |l, m\rangle.$$

(i) Show that the square of the eigenvalue of L_z cannot exceed of the eigenvalue of $\mathbf{L}^2$. [2 points]

(ii) Calculate $\langle L_x \rangle$ and $\langle L_x^2 \rangle$ for the state $|l, m\rangle$. [3 points]

Hint: You may use the relations $L_\pm = L_x \pm i\, L_y$ *and* $L_\pm|l, m\rangle = \hbar\, \sqrt{l(l+1) - m(m \pm 1)}\, |l, m \pm 1\rangle$.

(b) Consider a quantum particle in a 3D-potential $V(\mathbf{r})$.

(i) Prove that the rate of change of the expectation value of the orbital angular momentum $\mathbf{L}$ is equal to the expectation value of the torque:

$$\frac{d}{dt}\langle \mathbf{L} \rangle = \langle \mathbf{N} \rangle,$$

where $\mathbf{N} = \mathbf{r} \times (-\nabla V)$. You may use the formula $i\hbar \frac{\partial V}{\partial x_i} = [V, p_i]$. [4 points]

(ii) Calculate $\frac{d}{dt}\langle \mathbf{L} \rangle$ for any spherically symmetric potential and explain you result. [1 point] 5

5. (a) Consider two non-interacting systems with angular momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$ with angular momenta $j_1 = \frac{1}{2}$ and $j_2 = \frac{1}{2}$ respectively. The combined system has the total angular momentum j and its z-component m. Here we used the total angular momentum operator as the sum of individual ones, i.e., $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$. One may express the angular momentum state (in coupled bases) $|j_1, j_2; j, m\rangle$ of the composite in terms of uncoupled bases $|j_1, j_2; m_1, m_2\rangle$. Here, m_1 and m_2 are the z-component of individual angular momenta $\mathbf{J}_1$ and $\mathbf{J}_1$, respectively, and m is z-component of total angular momentum $\mathbf{J}$.

$|1,0\rangle$ 10

Find the matrix representation of the angular momentum operators J_x, J_y, and J_z in the coupled bases $|j_1, j_2; j, m\rangle$. (2+2+1 points)

(b) A particle of mass m is constrained to move between two concentric impermeable spheres of radii $r = a$ and $r = b$, with $b > a$. There is no other potential. Find the ground state energy and normalized wave function. [5 points]